

Chapter 4: Describing Data: Displaying and Exploring Data

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Introduction

In this chapter, we are interested in illustrating our numerical ways of measuring a dataset.

We will learn about several visualisation methods, such as the dot plot, box plot, scatter diagrams and the correlation coefficient.

Dot Plot

Consider the number of cars serviced by a Toyota and Lexus showroom for each working day of September 2026:

Toyota Showroom

Day	Cars Serviced
1	23
2	33
3	27
4	28
5	39
6	26
⋮	
24	30

Lexus Showroom

Day	Cars Serviced
1	31
2	35
3	44
4	36
5	34
6	37
⋮	
24	33

Dot Plot

We are interested in illustrating this data in a way that highlights its distribution. One way to do this is by using a **dot plot**.

This is constructed by placing each observation on a horizontal number line.

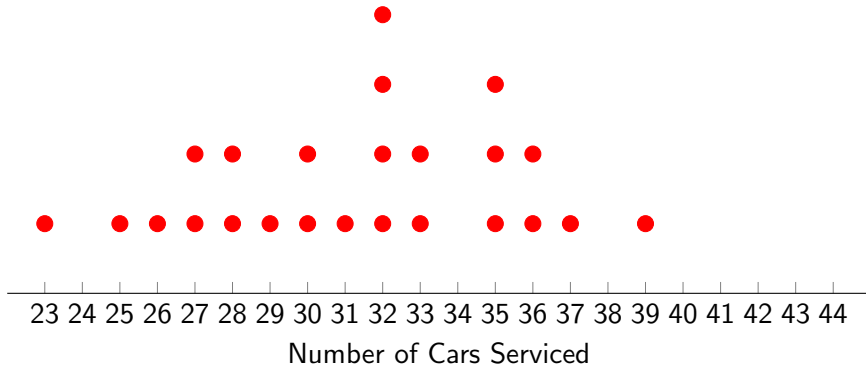
A dot is placed above the corresponding value for each observation in the dataset.

If a value occurs more than once, the dots are stacked vertically, so the height of each stack represents the frequency of that value.

Dot Plot

For the Toyota showroom:

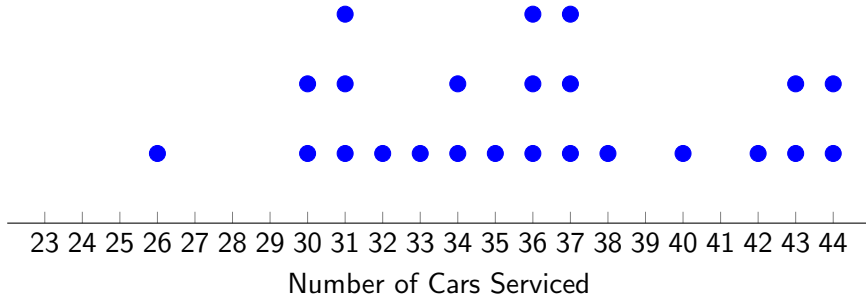
Figure: Toyota Showroom



Dot Plot

For the Lexus showroom:

Figure: Lexus Showroom



Dot Plot

The dot plots suggest that the Lexus showroom generally services more cars per day than the Toyota showroom, as its distribution is centred further to the right.

The Lexus data is also somewhat more dispersed, indicating greater day-to-day variation in the number of cars serviced.

Dot Plot

We can also analyse this numerically using some descriptive statistics:

Table: Descriptive Statistics

Showroom	n	Mean	Variance	St. Deviation	Median	Min	Max
Toyota	24	31.29	16.91	4.11	32	23	39
Lexus	24	35.83	24.58	4.96	36	26	44

The visualisation is supported by the numerical descriptives. The Lexus showroom has a larger average as well as a wider variance and standard deviation.

Measures of Position

Consider a small ordered dataset:

10, 15, 20, 25, 30, 35, 40

Suppose we are interested in finding the median.

The data is already arranged from lowest to highest and there are 7 observations. The median is therefore 25.

Measures of Position

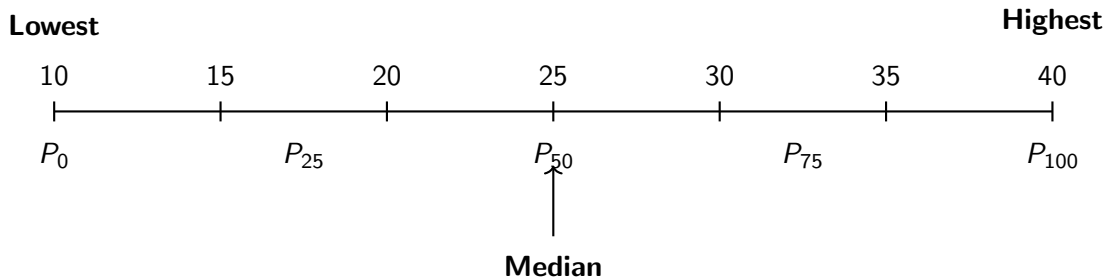
The median describes a position within the distribution.

More specifically, we say that the median is **the 50th percentile**.

This is because from its position, approximately 50% of the observations lie below it.

Measures of Position

So if the median identifies the point with 50% of the data, then we can generalise this to other percentiles, such as 25% or 75% of our data.



Measures of Position

More generally, the k th percentile P_k is the data value that indicates the position below which approximately $k\%$ of observations lie below it.

For example, the data value of 40 is at the 100th percentile. This means 100% of observations are at or below it.

Measures of Position

To identify the location L of the k th percentile p , we use the below formula:

$$L_p = \frac{k}{100}(n + 1)$$

Activity

You have a dataset of salespeople from the Toyota showroom and you record their commission received from their car sales. Identify the 25th, 50th and 75th percentile.

Salesperson	Commission (BD)
1	1,460
2	1,471
3	1,637
4	1,721
5	1,758
6	1,787
7	1,940
8	2,038
9	2,047
10	2,054
11	2,097
12	2,205
13	2,287
14	2,311
15	2,406

Measures of Position

The data is already sorted from smallest to largest. To identify the location, we use the below formula:

$$L_p = \frac{k}{100}(n + 1)$$

Measures of Position

There are $n = 15$ observations, which means for each percentile P_{25} , P_{50} and P_{75} :

$$L_{25} = \frac{25}{100}(15 + 1) = 4\text{th Observation}$$

$$L_{50} = \frac{50}{100}(15 + 1) = 8\text{th Observation}$$

$$L_{75} = \frac{75}{100}(15 + 1) = 12\text{th Observation}$$

Measures of Position

These identify the positions. Therefore:

$$P_{25} = 1721$$

$$P_{50} = 2038$$

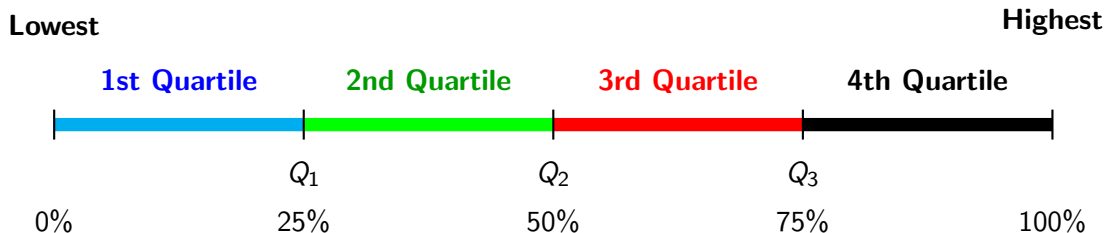
$$P_{75} = 2205$$

Measures of Position

A *percentile* divides the data into 100 equal parts. That is: $P_1, P_2, P_3, \dots, P_{100}$.

You should also be aware that we can also express this in **quartiles**. This means we divide the dataset into 4 equal parts:

Measures of Position



Note that $Q_1 = P_{25}$, $Q_2 = P_{50}$ and $Q_3 = P_{75}$. Notice that even though we label 3 quartiles, the data is divided into 4 parts.

Measures of Position

You should know that an interesting descriptive statistics is the **Interquartile Range** (IQR). This is simply the values where:

$$IQR = Q_3 - Q_1$$

Activity

Consider again our data on sales commissions:

Salesperson	Commission (BD)
1	1,460
2	1,471
3	1,637
4	1,721
5	1,758
6	1,787
7	1,940
8	2,038
9	2,047
10	2,054
11	2,097
12	2,205
13	2,287
14	2,311
15	2,406

Identify the Q_1 , Q_2 , Q_3 and derive the IQR .

Measures of Position

Solution:

Remember that $Q_1 = P_{25}$, $Q_2 = P_{50}$ and $Q_3 = P_{75}$. There are also $n = 15$ observations. Essentially, the answer is unchanged compared to the previous activity:

$$L_{25} = \frac{25}{100}(15 + 1) = 4\text{th Observation}$$

$$L_{50} = \frac{50}{100}(15 + 1) = 8\text{th Observation}$$

$$L_{75} = \frac{75}{100}(15 + 1) = 12\text{th Observation}$$

Measures of Position

These identify the positions. Therefore:

$$Q_1 = P_{25} = 1721$$

$$Q_2 = P_{50} = 2038$$

$$Q_3 = P_{75} = 2205$$

The IQR is therefore:

$$IQR = 2205 - 1721 = 484$$

Box Plot

We can visualise the quartiles of a dataset by illustrating it into a box plot. This will allow us to better understand the distribution of a dataset.

In order to construct a box plot, you need the following details:

- Q_1 .
- Q_2 .
- Q_3 .
- The maximum value of the dataset.
- The minimum value of the dataset.

Box Plot

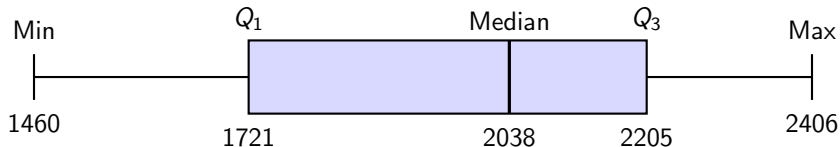
From our previous activity we can identify this:

- $Q_1 = 1721$.
- $Q_2 = 2038$.
- $Q_3 = 2205$.
- $\max = 2406$.
- $\min = 1460$.

Box Plot

We then draw the box plot as follows:

Figure: Box Plot



Scatter Diagram

A scatter diagram is usually used when trying to understand the correlation of two variables.

This is when we have two quantitative variables, one listed on the vertical axis and the other on the horizontal axis.

Scatter Diagram

Consider this sample dataset of 9 classes in the Quantitative Business Methods module of Bahrain Polytechnic. We collect data on the following two variables:

Table: Class Characteristics and Mathematics Scores

School	Class Size	Teacher Salary (BD)	Math Score
Class 1	18	1,480	88
Class 2	21	1,350	84
Class 3	23	1,420	86
Class 4	25	1,180	78
Class 5	27	1,250	80
Class 6	30	1,050	73
Class 7	32	1,120	75
Class 8	35	950	68
Class 9	38	880	64

Scatter Diagram

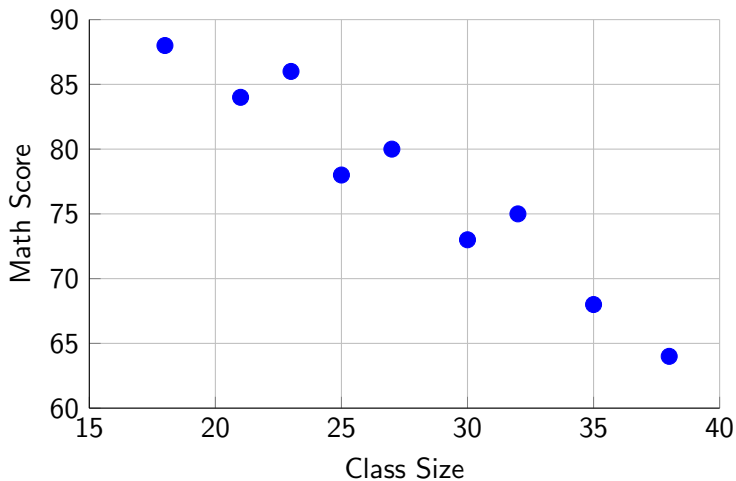
Suppose we are interested in the relationship between the size of a classroom and the average math score.

One way to see this is to have the values of each variable on the axes.

For example, lets have the math score on the vertical axis and the class size on the horizontal axis:

Scatter Diagram

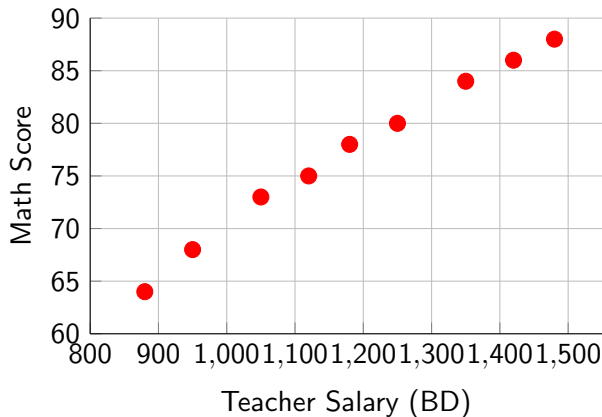
Figure: Class Size and Mathematics Scores



Scatter Diagram

Similarly, let us say we want to plot the relationship between math scores and the salary of the teacher leading the class:

Figure: Teacher Salary and Mathematics Scores

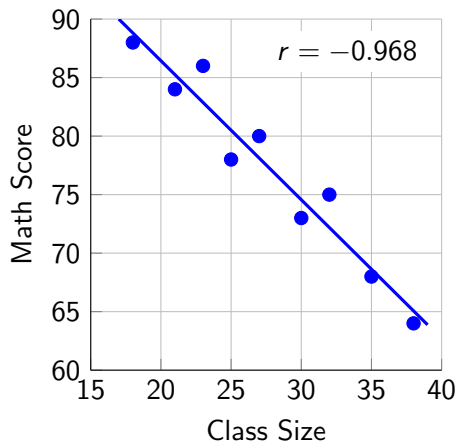


Correlation Coefficient

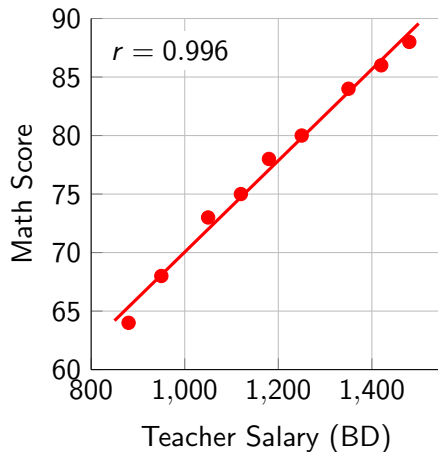
When variables are related to each other, as they are clearly shown in the scatter plot, then we can express this visually using a correlation graph:

Correlation Coefficient

Figure: Relationships between School Characteristics and Mathematics Scores



(a) Class Size and Math Score



(b) Teacher Salary and Math Score

Correlation Coefficient

The correlation coefficient r is a number that ranges between -1 and +1.

- When the correlation r is very close to -1, we say the variables are **strongly** negatively correlated.
- When the correlation r is very close to +1, then we say it is **strongly** positively correlated.
- When the correlation r is close to 0, then we say there is no relationship.

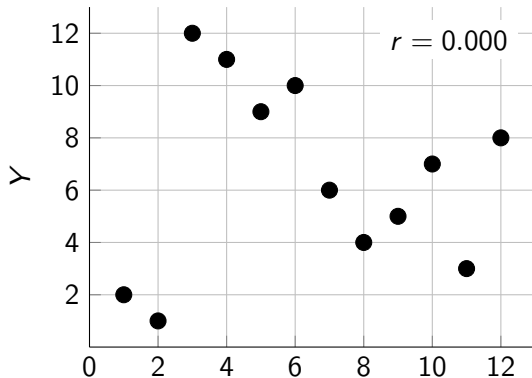
Correlation Coefficient

We can see in the plot that $r = -0.968$ for math and class sizes whereas it is $r = 0.996$ for the math and teacher salary. This is what we expected from our analysis earlier.

Correlation Coefficient

Activity:

Consider the below scatter plot between variables Y and X . What is the relationship between the two variables?



Correlation Coefficient

There is no relationship between Y and X .

This is seen from the scatter plot itself which does not consistently show any positive or negative relation.

Moreover, the $r = 0$, so there is no correlation between them.

Contingency Table

A scatter plot is useful when analysing the relationship between two quantitative variables.

However, if we are interested in the relationship between **categorical** variables, then we could consider using a **contingency table**.

Contingency Table

Suppose the CEO of Batelco wants to know about the performance of their branches across Bahrain. Data on customer satisfaction and branch location is given below:

Table: Customer Satisfaction by Location

Satisfaction	Manama	Riffa	Muharraq	Isa Town	Total
Satisfied	25	20	19	26	90
Dissatisfied	27	20	26	17	90
Total	52	40	45	43	180

Contingency Table

It seems that across all branches, customers are equally happy as they are unhappy. We can see this by looking at the last column under **Total**:

$$\frac{\text{Satisfied}}{\text{Total}} = \frac{90}{180} = 0.5$$

therefore, 50% of all customers surveyed are happy with their service across all branches.

Contingency Table

Activity:

Between each branch, which location has the best customer satisfaction?

Contingency Table

Solution:

You need to divide the satisfaction of each branch by their total. We can see that the Isa Town branch has the highest satisfaction:

$$\frac{26}{43} = 0.64$$

meaning that 60.4% of customers were happy with the service.